

Website design evaluation: reward, metrics, and agent behavior

We built a browser-state evaluator for screenshot-to-code website reproduction. The main finding is clear: broad visual similarity is not enough. The reward needs rendered HTML, CSSOM, visual blocks, and state coverage to distinguish faithful reproduction from a plausible page in the same genre.

12

synthetic websites

Multi-page tasks generated as a first dataset for Harbor evaluation.

153

capture rows

Full-page and stateful captures: dropdowns, filters, tabs, accordions, selected nodes.

0.194-0.678

reward spread

Enough separation to inspect what the reward is actually learning.

Core insight

DreamSim and global pixelmatch often stay high for weak candidates. The metrics that separate good from bad are rendered HTML, CSSOM, visual-block geometry, bbox layout, and state coverage.

1. What we built

The evaluator treats the browser as the source of truth. For each oracle capture, it serves the reference and candidate pages, replays the capture state, then extracts screenshots, rendered DOM/outerHTML, CSSOM, visual blocks, and layout boxes from that live page state.

This matters because future candidates may be plain HTML, React, Solid, Tailwind, or a build system. The reward should not care about source shape. It should care about what the browser renders.

Design principle

The oracle manifest defines what is fair to score. If a dropdown, filter, tab, or selected state is in the oracle manifest, then we evaluate whether the candidate can reproduce that visible state.

Pipeline

1. Generate synthetic oracle website.
2. Capture oracle screenshots and state manifest.
3. Give only screenshots to the coding agent.
4. Agent builds candidate website.
5. Evaluator generates candidate capture plan and replays states.
6. Reward combines coverage, visual, DOM, CSSOM, VLM, and geometry metrics.

The run was analytically valid: metrics completed for all 12 trials. Some trials hit an old markdown report writer bug after metrics completed; that bug does not affect the score data used here.

2. Reward overview

The scalar reward is a weighted score per manifest capture. Coverage is applied once as the outer multiplier. Unsupported evaluator components are removed from the available denominator; numeric zeros remain real zeros.

Component	Weight	Role
Screenshot size match	0.05	Cheap rendering/page-length guard.
Rendered HTML	0.10	Content and DOM structure.
VLM	0.20	Human-like visual judgement.
Pixel match	0.05	Global + blockwise pixel fidelity.
Visual block aggregate	0.00	Reported as evidence; not directly weighted in the current scalar reward.
BBox geometry	0.10	Block shape and placement, derived from visual-block matching.
CSSOM style	0.10	Computed style fidelity over matched visual blocks.
DreamSim	0.10	Broad perceptual similarity.

Gate

A capture must pass screenshot size, rendered HTML, and VLM thresholds before the more specific components contribute. This prevents a page with the wrong content or broken rendering from receiving high credit because one visual metric is high.

3. Trial-level reward results

The spread is meaningful: the best candidate is about 3.5x the lowest reward. DreamSim remains high for weak candidates, while rendered HTML, CSSOM, and visual-block/geometry metrics separate fidelity.

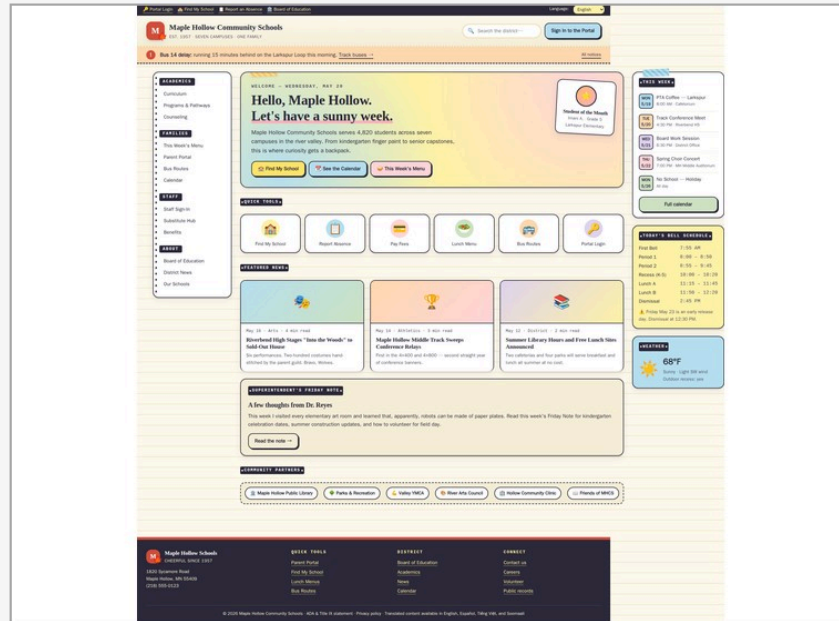
Trial	Reward	Coverage	HTML	CSSOM	VB	Read
site-01-education-k12	0.678	0.957	0.634	0.646	0.855	Best overall. Structure, content, and styling line up.
site-05-retail-marketplace	0.620	0.882	0.507	0.589	0.767	Good visual system; missing density states.
botanical-garden-nursery	0.607	0.975	0.477	0.622	0.831	Strong visual blocks; one state miss.
site-04-science-research	0.599	0.959	0.496	0.452	0.750	Good broad reproduction; weaker style/geometry.
podcast-network	0.596	0.944	0.473	0.509	0.759	Good page vibe; state gaps remain.
storm-chasers-community	0.576	0.882	0.541	0.519	0.748	Decent static structure.
regional-credit-union	0.569	1.000	0.485	0.521	0.809	Coverage complete; local branch-state geometry failed.
tabletop-gaming-cafe	0.556	0.850	0.496	0.494	0.711	Low coverage from missed interaction states.
zen-center	0.548	0.943	0.438	0.575	0.801	High broad visual scores; HTML gate catches drift.
curling-federation	0.414	0.947	0.382	0.503	0.776	Same domain, wrong density/card system.
indie-record-label	0.261	1.000	0.314	0.420	0.768	Plausible vibe, weak rendered content structure.
lutherie-guild	0.194	1.000	0.279	0.271	0.699	DreamSim high, real structure/style low.

4. Evidence: high reward requires page grammar, not just resemblance

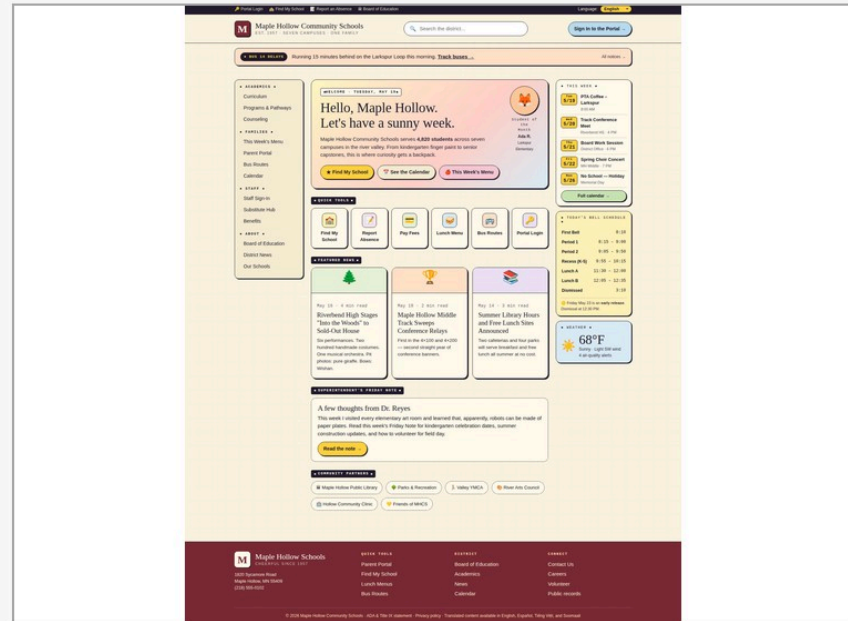
Hypothesis 1: High reward when page grammar and rendered structure line up

The education task scores highest because the candidate keeps the dashboard layout, side rail, dense cards, school palette, and much of the rendered content structure.

Reference / oracle



Candidate



capture score: 0.735 | DreamSim: 0.847 | VLM: 0.700 | HTML: 0.655 | CSSOM: 0.805 | VB: 0.919 | VB size: 0.994 | VB text: 0.927 | block PM: 0.353 | BBox: 0.428

K-12 school site: strongest overall candidate. It preserves dashboard-like layout, side rail, dense cards, palette, footer, and enough rendered content structure.

The highest-scoring task is not merely attractive. It keeps the page grammar: repeated structures, content density, navigation, and styling survive the reproduction.

reward 0.678

HTML 0.634

CSSOM 0.646

visual block 0.855

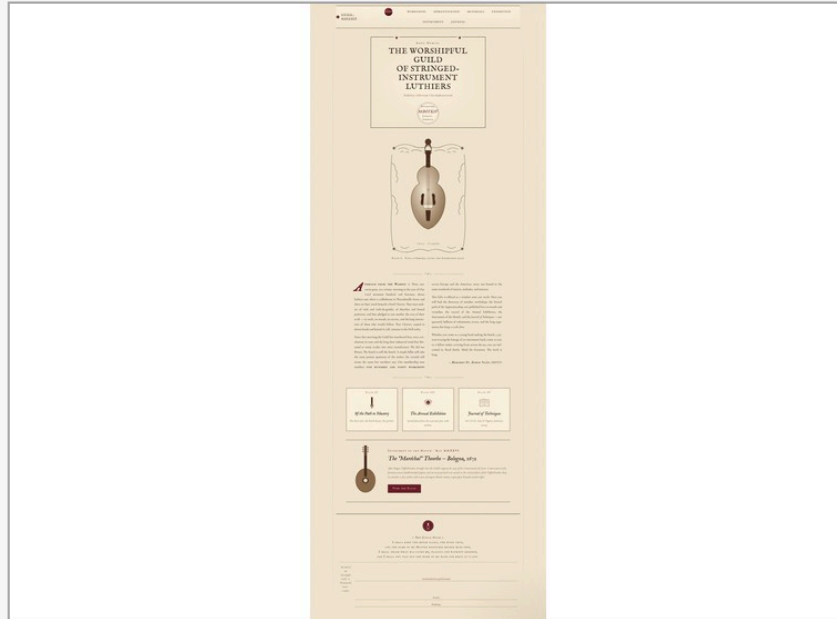
coverage 0.957

5. Evidence: DreamSim over-credits broad aesthetic resemblance

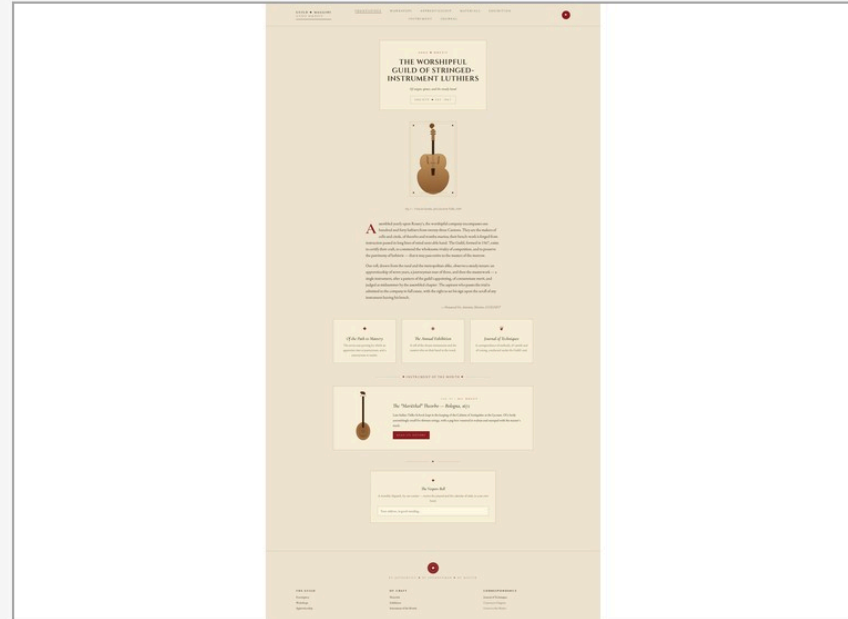
Hypothesis 2: DreamSim can over-credit broad aesthetic resemblance

The lutherie candidate preserves the parchment/minimal mood, so broad perceptual similarity is not catastrophic, but local structure, text, CSSOM, and bbox fidelity are poor.

Reference / oracle



Candidate



capture score: 0.181 | DreamSim: 0.846 | VLM: 0.521 | HTML: 0.238 | CSSOM: 0.200 | VB: 0.698 | VB size: 0.233 | VB text: 0.825 | block PM: 0.144 | BBox: 0.104

Lutherie guild: parchment mood and central composition survive, but actual structure, typography, copy, and layout fidelity are poor.

This is the cleanest warning against making DreamSim a primary reward driver. The broad aesthetic family is right, but the reproduction is not faithful.

reward 0.194

DreamSim 0.849

HTML 0.279

CSSOM 0.271

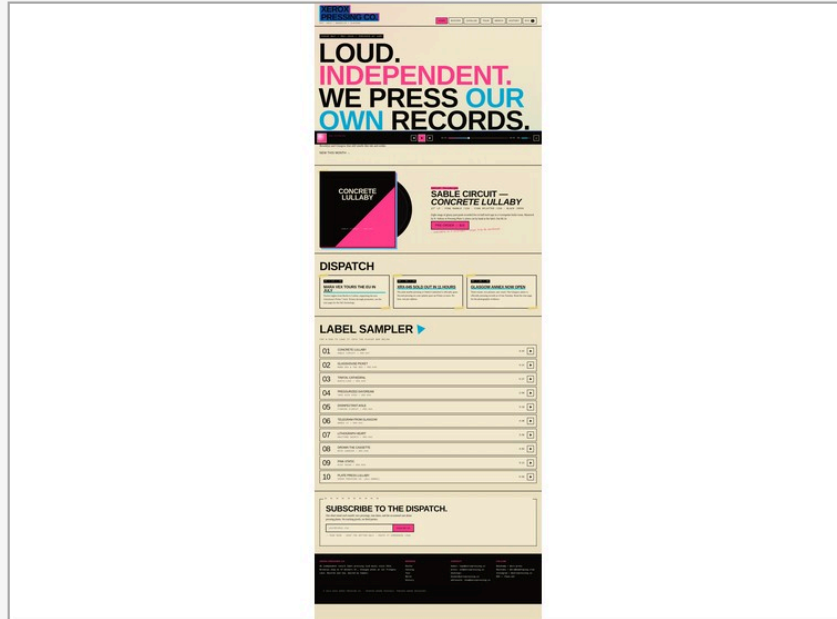
BBox 0.143

6. Evidence: plausible visual clone, weak rendered structure

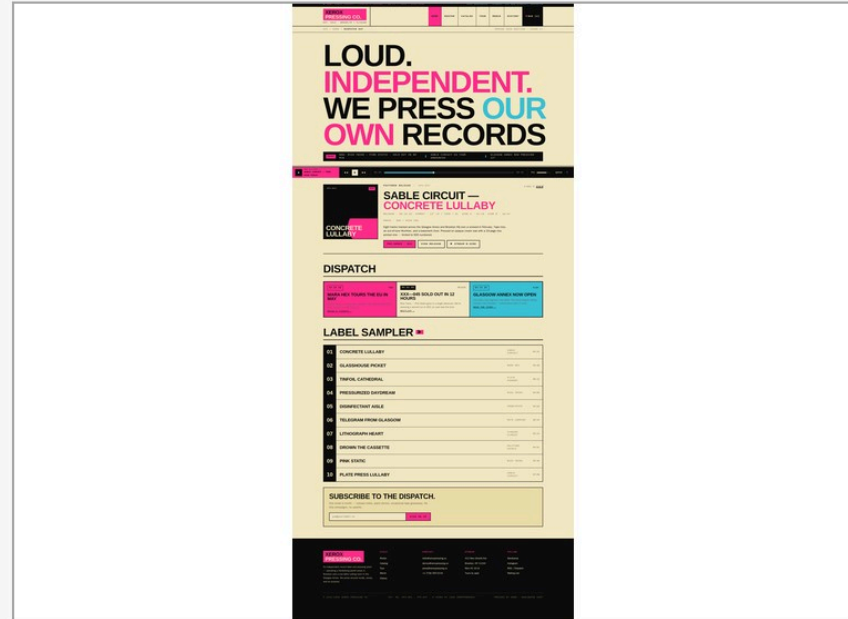
Hypothesis 3: A plausible visual clone can still miss rendered HTML/content structure

The indie home page looks recognizably close at first glance, but the reward falls because rendered text/content and local CSS structure drift across captures.

Reference / oracle



Candidate



capture score: 0.250 | DreamSim: 0.844 | VLM: 0.744 | HTML: 0.354 | CSSOM: 0.619 | VB: 0.810 | VB size: 0.760 | VB text: 0.895 | block PM: 0.481 | BBox: 0.498

Indie record label: the candidate has the right vibe, but content structure and page-specific details drift.

A judge or perceptual model can be tempted by a plausible page. Rendered HTML and CSSOM prevent over-crediting by checking whether the actual browser-rendered structure is close.

reward 0.261

VLM 0.652

visual block 0.768

HTML 0.314

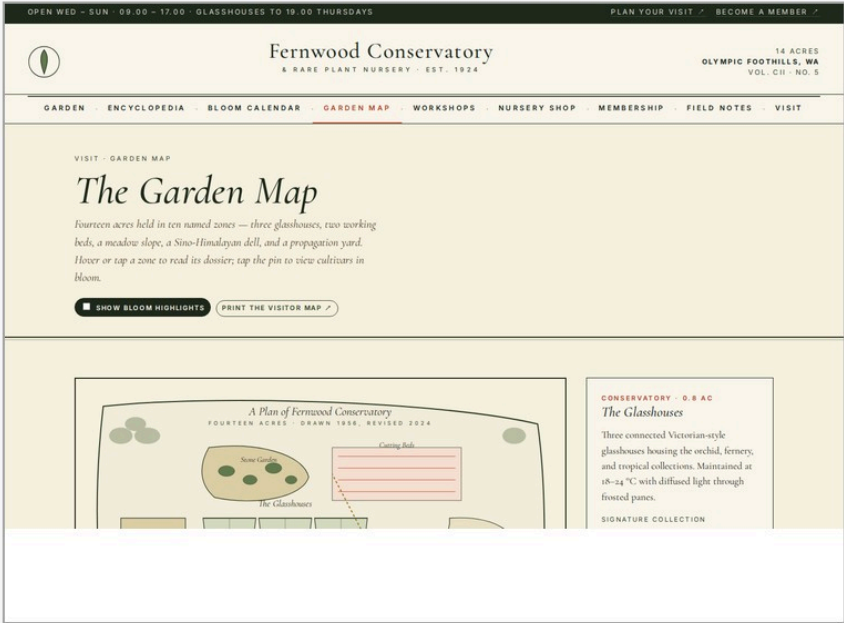
CSSOM 0.420

7. Evidence: visual block works when visible blocks align

Hypothesis 4: Visual block works well for stateful local layout when blocks align

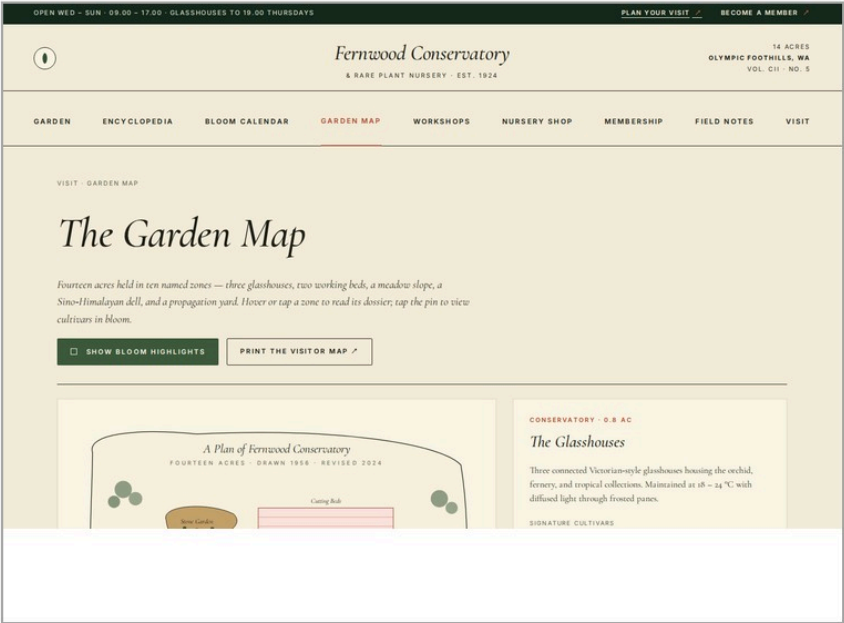
Botanical garden-map state has high visual-block, block pixelmatch, CSSOM, and final capture score because the same visible block system is largely present.

Reference / oracle



capture score: 0.810 | DreamSim: 0.885 | VLM: 0.842 | HTML: 0.592 | CSSOM: 0.863 | VB: 0.951 | VB size: 0.992 | VB text: 0.953 | block PM: 0.606 | BBox: 0.482

Candidate



Botanical garden map state: labels, cards, map area, and highlighted state structure mostly align.

This is the intended use case for visual block scoring. Once visible blocks correspond, the metric checks whether the same pieces exist with similar area, text, placement, and style.

capture score 0.810

VB 0.951

VB size 0.992

block PM 0.606

CSSOM 0.863

8. Evidence: text can match while geometry is wrong

Hypothesis: text can match while state geometry is wrong

Same booking/contact capture. The candidate has much of the contact-form text, but the layout/state is wrong: previous step content remains above it and the reservation panel geometry changes.

Reference / oracle

Meeple & Mug

BOARD GAME CAFE

Home

Menu

Library

Events

Book

Tavern Pass

Reviews

Book a table

1 Pick room

2 Date & party

3 Add-ons

4 Contact & deposit

Step 4 · Contact & deposit

Your name

Riley Tabletop

Phone

(555) 100-2000

Email

riley@example.com

Occasion (optional)

Friends night

Anything we should know?

Dietary needs, surprise guest, song requests...

☐ I agree to the deposit terms. The deposit applies to my final tab; refundable up to 48h before.

← Back

Pay deposit & reserve →

Your reservation

Room

— not selected —

When

2026-05-23 · 3h

Party

8 players

Estimated total

\$0

Deposit due

\$0

Deposit applies to your final tab. Free cancellation 48h ahead.

capture score 0.534 | HTML 0.613 | VB 0.656 | VB text 0.939 | VB size 0.061 | block PM 0.027 | BBox 0.026 | CSSOM 0.047

Candidate

Meeple & Mug

BOARD GAME CAFE

Home

Menu

Library

Events

Book

Tavern Pass

Reviews

Book a table

Step 3 · Add-ons (optional)

Tap to toggle. We'll add to the running total in the sidebar.

Snack platter

Hummus, fries, olives, pretzels — feeds 6. +\$24

Add

Dedicated host

A staff member explains rules, restocks dice, herds your group. +\$45

Add

Themed setup

We pre-arrange games on your theme: heist, fantasy, hidden ID. +\$15

Add

← Back

Next contact →

Step 4 · Contact & deposit

Your name

Riley Tabletop

Phone

(555) 100-2000

Email

riley@example.com

Occasion (optional)

Friends night

Tabletop booking contact state: the contact-form text is present, but the candidate keeps the previous add-ons step above it and changes the reservation-panel geometry.

This case explains why visual block cannot be reduced to text. The candidate includes much of the right form content, but the block area, local pixels, BBox, and CSSOM show that the state layout is wrong.

capture score 0.534

HTML 0.613

VB text 0.939

VB size 0.061

block PM 0.027

BBox 0.026

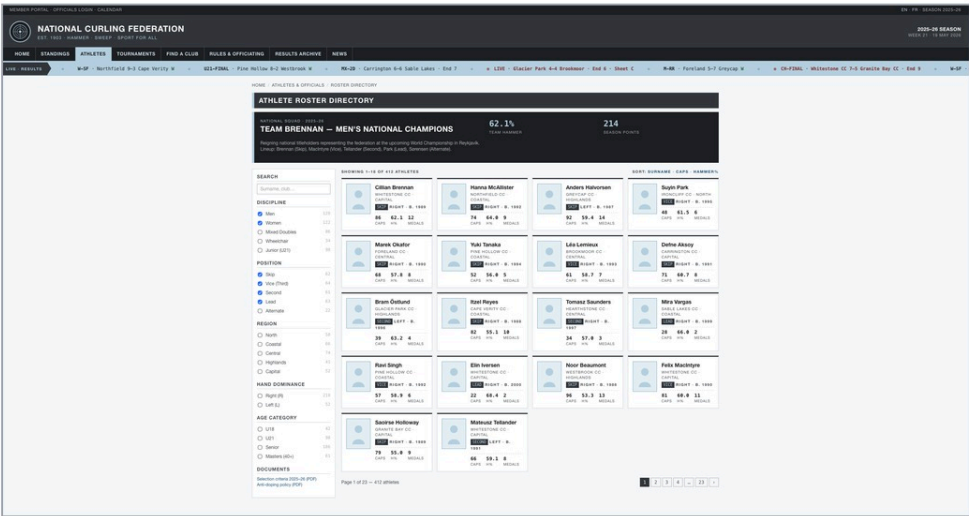
CSSOM 0.047

9. Evidence: local density matters

Provided evidence: curling athlete directory local mismatch

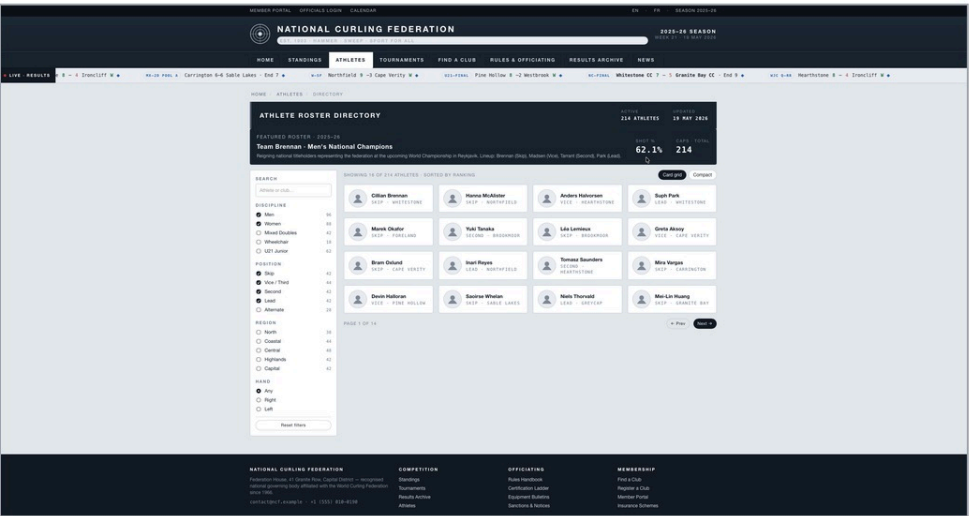
Corrected attribution: oracle/reference is the denser roster page; candidate is the compact page with footer.

Reference / oracle



visual conclusion: broad match, local density/card/header/footer mismatch

Candidate



Curling athlete directory: the page identity is correct, but card density, header geometry, filter/sidebar treatment, player count layout, and footer/page geometry differ.

This is the case for local fidelity metrics. Broad screenshot identity and visual-block aggregate can still recognize the page, but HTML, CSSOM, BBox, and blockwise pixelmatch correctly penalize density and layout drift.

reward 0.414

athletes capture 0.218

DreamSim 0.766

HTML 0.346

BBox 0.119

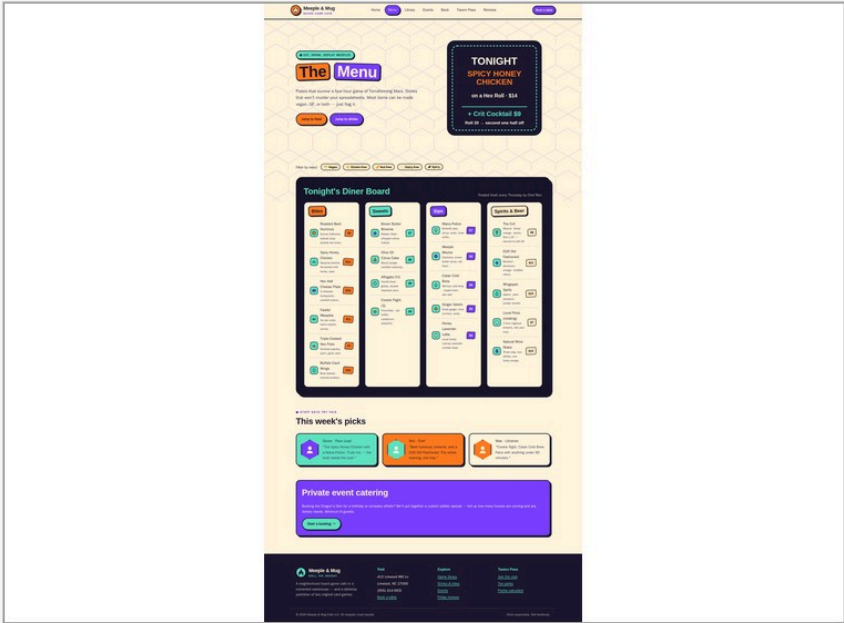
CSSOM 0.369

10. Evidence: agents reproduce vibe better than states

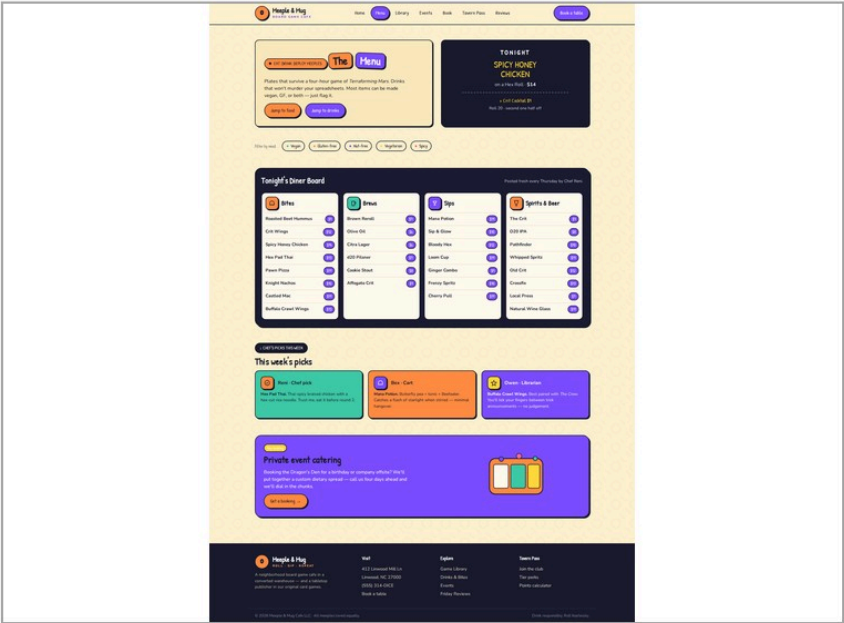
Hypothesis 7: Agents reproduce page vibe more reliably than interaction/state captures

The tabletop candidate reproduces the colorful board-game cafe vibe, but several requested states were missing or not visually active; base pages can look good while state coverage remains weak.

Reference / oracle



Candidate



capture score: 0.649 | DreamSim: 0.880 | VLM: 0.654 | HTML: 0.505 | CSSOM: 0.635 | VB: 0.820 | VB size: 0.731 | VB text: 0.912 | block PM: 0.298 | BBox: 0.247

Tabletop cafe: cheerful domain style is present, but multiple requested state captures are missing.

The agent is stronger at static page construction than interaction-state reproduction. Missing filters, toggles, accordions, selected nodes, and alternate views should remain first-class reward failures.

reward 0.556

coverage 0.850

missing 4/17 states

visual style plausible

11. Metric usefulness

Metric	Usefulness	Evidence	How to use it
Rendered HTML	Very strong	Trial corr 0.933	Best discriminator for content, section, and repeated DOM structure drift.
CSSOM	Very strong	Trial corr 0.840	Catches browser-computed style: typography, spacing, color, layout.
Visual block aggregate	Strong	Capture corr 0.587	Useful visible-block fidelity signal, but must be read with submetrics.
Visual block size	Strong	Capture corr 0.558	Tells whether the same visual block area exists. This catches missing/reshaped components.
Visual block text	Strong but incomplete	Capture corr 0.516	Good content-block signal; can overcredit when text appears in the wrong geometry.
Blockwise pixelmatch	Useful local check	Capture corr 0.407	Compares pixels inside matched blocks, avoiding many whole-page false positives.
BBox geometry	Useful but harsh	Capture corr 0.456	Good for card density and placement; not reliable alone.
VLM judge	Useful sanity signal	Trial corr 0.409	Sees human-facing quality, but can overrate plausible wrong structure.
DreamSim	Broad only	Compressed 0.747-0.862	Good catastrophic mismatch guard, weak final reward driver.
Global pixelmatch	Debug only	Trial corr ~0	Too sensitive to whitespace/page size and too forgiving when backgrounds dominate.
Masked CLIP	Do not use yet	Skipped/null	No evidence it adds useful signal; avoid CLIP forward passes for now.

Important

DreamSim and global pixelmatch are not wrong metrics; they are wrong as dominant reward drivers. They are useful as guards or diagnostics, while rendered HTML/CSSOM/visual block carry the real fidelity signal.

12. Visual block breakdown

Visual block is useful because it decomposes visible text/content blocks. The aggregate score is less informative than the submetrics.

Submetric	Range	Mean	Capture-score corr	Interpretation
Size	0.038-0.997	0.668	0.558	Whether the same visible block area exists.
Text	0.697-1.000	0.908	0.516	Whether matched blocks contain similar text.
Position	0.254-0.966	0.861	0.313	Whether the block lands in a similar location.
Text color	0.731-0.968	0.845	0.158	Mostly diagnostic; weak reward driver.
Block pixelmatch	0.006-0.713	0.316	0.407	Local pixel fidelity inside matched blocks.

What worked

Size and text were the strongest visual-block submetrics. Size catches whether the same visible block area exists; text catches whether the block content is similar.

What is niche

Text color is useful for diagnosis but weak as a reward driver. Blockwise pixelmatch is more useful than global pixelmatch because it focuses on matched local components.

13. Agent insights and reward implications

What the agent does well

- Creates plausible page scaffolds and routes.
- Captures broad art direction: palette, mood, hero/footer, domain feel.
- Builds recognizable components such as cards, grids, sidebars, dashboards, menus, and editorial sections.

Where it struggles

- Exact card density, spacing, row/column geometry, and local alignment.
- Rendered text/content structure and repeated DOM patterns.
- Stateful captures: filters, accordions, selected nodes, tabs, toggles, compare states.
- Custom diagrams/assets such as maps, brackets, rosters, botanical drawings, and audio panels.

Reward changes

- Keep rendered HTML and CSSOM central.
- Keep visual block, but always report size/text/position/color/block-pixelmatch.
- Use blockwise pixelmatch, not global pixelmatch, as the local pixel signal.
- Keep VLM as a sanity signal, paired with deterministic browser-rendered artifacts.
- Keep DreamSim small: good guard, weak fine-grained reward.
- Treat state misses as first-class reward events, not evaluator noise.

Final read

The reward is directionally working because it does not simply reward broad visual similarity. Weak candidates can look plausible, but final reward drops when rendered HTML, CSSOM, visual block size, bbox geometry, or state coverage fail.